

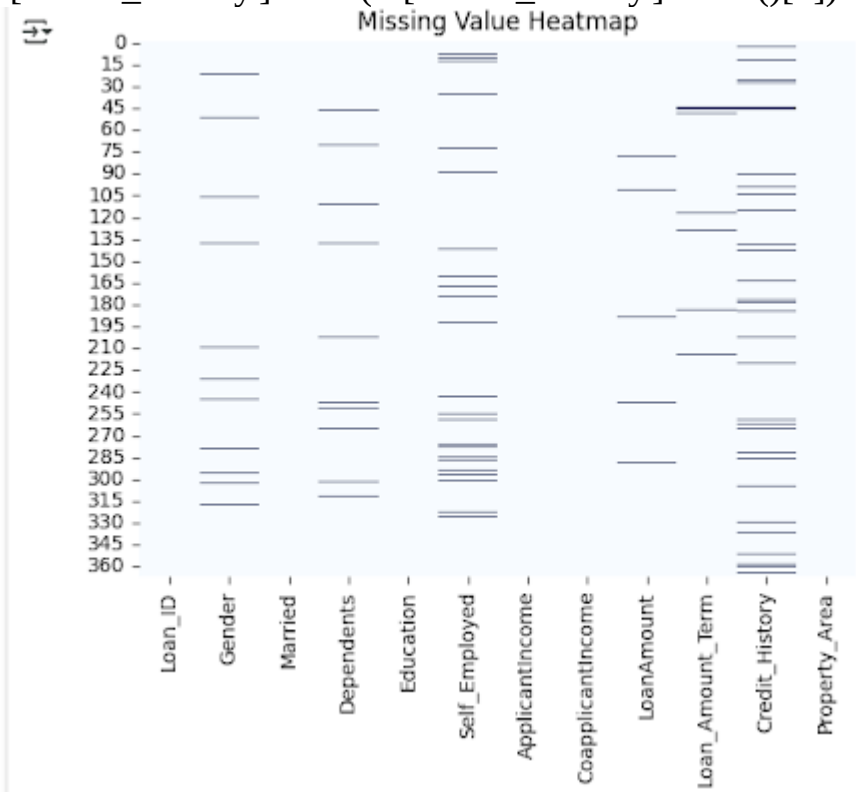
- Handling missing values: detection, filling, and dropping
- Removing duplicates and unnecessary data
- Data type conversion and ensuring consistency
- Normalize data (e.g., standardization, min-max scaling).

CODE:

```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.preprocessing import MinMaxScaler, StandardScaler
df = pd.read_csv('/content/test_Y3wMUE5_7gLdaTN.csv')
print("Initial Data Overview:")
print(df.info())
```

```
Initial Data Overview:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 367 entries, 0 to 366
Data columns (total 12 columns):
 #   Column                Non-Null Count  Dtype
---  -
 0   Loan_ID               367 non-null   object
 1   Gender                356 non-null   object
 2   Married               367 non-null   object
 3   Dependents            357 non-null   object
 4   Education             367 non-null   object
 5   Self_Employed         344 non-null   object
 6   ApplicantIncome       367 non-null   int64
 7   CoapplicantIncome     367 non-null   int64
 8   LoanAmount            362 non-null   float64
 9   Loan_Amount_Term      361 non-null   float64
10   Credit_History         338 non-null   float64
11   Property_Area         367 non-null   object
dtypes: float64(3), int64(2), object(7)
memory usage: 34.5+ KB
None
```

```
print("\nMissing Values in Each Column:\n", df.isnull().sum())
sns.heatmap(df.isnull(), cbar=False, cmap="Blues")
plt.title("Missing Value Heatmap")
plt.show()
for col in ['Gender', 'Married', 'Dependents', 'Self_Employed']:
    df[col].fillna(df[col].mode()[0])
df['LoanAmount'].fillna(df['LoanAmount'].median())
df['Loan_Amount_Term'].fillna(df['Loan_Amount_Term'].mode()[0])
df['Credit_History'].fillna(df['Credit_History'].mode()[0])
```



Credit_History	
0	1.0
1	1.0
2	1.0
3	1.0
4	1.0
...	...
362	1.0
363	1.0
364	1.0
365	1.0
366	1.0

367 rows × 1 columns

dtype: float64

```
initial_rows = df.shape[0]
df.drop_duplicates(inplace=True)
print(f"\nRemoved {initial_rows - df.shape[0]} duplicate rows.")
```



```
Removed 0 duplicate rows.
```

```
[6]
```

```
df['Dependents'] = df['Dependents'].replace('3+', 3).fillna(0).astype(int)
for col in ['Gender', 'Married', 'Education', 'Self_Employed', 'Property_Area']:
    df[col] = df[col].str.strip().str.capitalize()
min_max_scaler = MinMaxScaler()
scale_cols = ['ApplicantIncome', 'CoapplicantIncome', 'LoanAmount']
df[scale_cols] = min_max_scaler.fit_transform(df[scale_cols])
scaler = StandardScaler()
df[['Credit_History']] = scaler.fit_transform(df[['Credit_History']])
print("\nCleaned Data Summary:")
print(df.info())
print(df.head())
```



```
Cleaned Data Summary:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 367 entries, 0 to 366
Data columns (total 12 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Loan_ID                               367 non-null    object
1   Gender                                356 non-null    object
2   Married                               367 non-null    object
3   Dependents                            367 non-null    int64
4   Education                             367 non-null    object
5   Self_Employed                         344 non-null    object
6   ApplicantIncome                       367 non-null    float64
7   CoapplicantIncome                     367 non-null    float64
8   LoanAmount                           362 non-null    float64
9   Loan_Amount_Term                       361 non-null    float64
10  Credit_History                        338 non-null    float64
11  Property_Area                         367 non-null    object
dtypes: float64(5), int64(1), object(6)
memory usage: 34.5+ KB
None
```

	Loan_ID	Gender	Married	Dependents	Education	Self_Employed	\
0	LP001015	Male	Yes	0	Graduate	No	
1	LP001022	Male	Yes	1	Graduate	No	
2	LP001031	Male	Yes	2	Graduate	No	
3	LP001035	Male	Yes	2	Graduate	No	
4	LP001051	Male	No	0	Not graduate	No	

	ApplicantIncome	CoapplicantIncome	LoanAmount	Loan_Amount_Term	\
0	0.078865	0.000000	0.157088	360.0	
1	0.042411	0.062500	0.187739	360.0	
2	0.068938	0.075000	0.344828	360.0	
3	0.032263	0.106083	0.137931	360.0	
4	0.045168	0.000000	0.095785	360.0	

	Credit_History	Property_Area
0	0.459858	Urban
1	0.459858	Urban
2	0.459858	Urban
3	NaN	Urban
4	0.459858	Urban